

# Alec Helbling

## *Interpretable and Controllable Generative Models*

I develop novel interfaces for controlling Generative AI. In particular, I leverage a combination of machine learning and interactive visualization to allow humans to control generative models.

I have worked with researchers and engineers at Georgia Tech, Apple, Adobe, IBM Research, NASA Jet Propulsion Laboratory, Microsoft, and the University of Pittsburgh.

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📄 CV PDF

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🔗 @helblazer811

🎓 Google Scholar

## Education

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Aug. 2023 -  
Present

### **Ph.D. in Machine Learning**

Georgia Institute of Technology, Atlanta, GA

Advisor: Duen Horng (Polo) Chau

🏆 NSF Fellowship, Georgia Tech President's Fellowship

Aug. 2018 - Dec.  
2022

### **B.S. in Computer Science**

Georgia Institute of Technology, Atlanta, GA

Overall GPA: 3.91/4.00, Highest Honors, Threads in Intelligence and Theory

## Industry Research Experience

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Summer 2025

### **Apple, Seattle, WA**

*AI Research Intern, Responsible AI Team*

Mentor: Joseph Cheng, Leon Gatys, Shruti Palaskar, Charlie Maalouf

I worked on improving the safety of vision language models.

Summer 2024 **Adobe Firefly**, San Jose, CA

*AI Research Intern*

Mentor: Siddharth Iyer, Oliver Brdiczka

I worked on improving the safety of large scale diffusion models.

Summer 2021, 2022 **IBM Research**, Yorktown Heights, NY

*AI Research Intern, Foundations of AI Reasoning Group*

Mentor: Achille Fokoue, Tengfei Ma, Maria Chang

I worked on systems for summarizing natural language documents in a way that preserves their factuality.

Summer 2020 **Microsoft**, Redmond, WA

*Software Engineering Intern, Microsoft Intune Insights and Analytics Group*

Mentor: Durga Kumar Varanasi

Developed a system for automatically distributing service traffic across several database instances.

Summer 2019 **NASA Jet Propulsion Lab**, Pasadena, CA

*Machine Learning Research Intern, Machine Learning and Instrument Autonomy Group*

Mentor: Lukas Mandrake, Jack Lightholder

I worked on interactive interfaces allowing mission scientists without ML expertise to deploy powerful ML methods.

## Academic Research Experience

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Aug. 2023 Present **Georgia Institute of Technology**, Atlanta, GA

*Graduate Research Assistant, School of Computational Science and Engineering*

Advisor: Duen Horng (Polo) Chau

I am a member of the Polo Club of Data Science. We focus on the intersection of machine learning, human computer interaction, and visualization.

Fall 2020–Spring 2023 **Georgia Institute of Technology**, Atlanta, GA

*Student Researcher, SIPLab*

Mentor: Chris Rozell, Kion Fallah, Matthew O'Shaughnessy

I worked on systems for controlling deep generative models for image generation by using simple pairwise feedback from people.

Mar. 2015 - Aug.  
2018

**University of Pittsburgh, School of Medicine, Pittsburgh, PA**

*Student Researcher and Javascript Programmer, Koes Lab*

Mentor: David Koes

I worked on methods for visualizing molecular structures in web browsers, applications of machine learning to computational drug discovery, and visualizations of machine learning models for drug discovery.

## Honors and Awards

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2024 **National Science Foundation Graduate Research Fellowship**

Prestigious fellowship with 3 years guaranteed funding.

2025 **Oral Presentation, ICML 2025**

For "ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features"

2025 **Best Paper Award, CVPR Workshop on Visual Concepts**

For "ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features"

2025 **Spotlight Oral Presentation, CVPR Workshop on Visual Concepts**

For "ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features"

2024 **Best Poster, IEEE VIS 2024**

For "Transformer Explainer - Interactive Learning of Text-Generative Models"

2023 **Best Poster, IEEE VIS 2023**

For "Manim ML: Communicating Machine Learning Architectures with Animation"

2023 — Present

**President's Fellowship at Georgia Institute of Technology**

Select number of 1st year doctoral students who bring exemplary levels of scholarship and innovation to their academic departments

2020, 2021 **President's Undergraduate Research Fellowship (x2)**

For "PrefGenML: Preference Guided Image Generation with Relative Attributes"

2018 Pittsburgh Science and Technology Academy Valedictorian  
Top of my high school class

## Publications

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### Selected

#### **ConceptAttention: Diffusion Transformers Learn Highly Interpretable Features**

Alec Helbling, Tuna Han Salih Meral, Benjamin Hoover, Pinar Yanardag, Duen Horng (Polo) Chau

*ICML, 2025*

🏆 Oral Presentation (top 1% of 12,000 submissions)

🏆 Best Paper Award CVPR 2025 Workshop on Visual Concepts

📄 PDF [</> Code \(354 ⭐\)](#)

#### **ManimML: Communicating Machine Learning Architectures with Animation**

Alec Helbling, Duen Horng (Polo) Chau

*IEEE VIS: Visualization and Visual Analytics, 2023*

🏆 Best Poster Award

📄 PDF [</> Code \(3254 ⭐\)](#)

#### **Diffusion Explorer: Interactive Exploration of Diffusion Models**

Alec Helbling, Duen Horng (Polo) Chau

*IEEE VIS, 2025*

📄 PDF [</> Code \(902 ⭐\)](#) [🔗 Tool](#)

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## All Publications

#### **SafetyPairs: Isolating Safety Critical Image Features with Counterfactual Image Generation**

Alec Helbling, Shruti Palaskar, Kundan Krishna, Polo Chau, Leon Gatys, Joseph Yitan Cheng

*arXiv preprint, 2025*

 [PDF](#)

### **Transformer Explainer: Interactive Learning of Text-Generative Models**

Aeree Cho (\*), Grace C Kim (\*), Alexander Karpekov (\*), Alec Helbling, Zijie J Wang, Seongmin Lee, Benjamin Hoover, Duen Horng Chau

*IEEE VIS: Visualization & Visual Analytics, 2024*

 **Best Poster Award**

 [PDF](#) [</> Code \(6051 !\[\]\(29658d981ebdf5edc259074cbf6110e0\_img.jpg\) \)](#)

### **UNDREAM: Bridging Differentiable Rendering and Photorealistic Simulation for End-to-End Adversarial Attacks**

Mansi Phute, Willian Lunardi, Matthew Hull, Haoran Wang, Martin Andreoni, Alec Helbling, Wenke Lee, ShengYun Peng, Polo Chau

*arXiv preprint, 2025*

 [PDF](#)

### **RenderBender: A Survey on Adversarial Attacks Using Differentiable Rendering**

Matthew Hull, Haoran Wang, Matthew Lau, Alec Helbling, Mansi Phute, Chao Zhang, Zsolt Kira, Willian Lunardi, Martin Andreoni, Wenke Lee, Polo Chau

*IJCAI Survey Track, 2024*

 [PDF](#)

### **Non-Robust Features are Not Always Useful in One-Class Classification**

Matthew Lau, Haoran Wang, Alec Helbling, Matthew Hull, ShengYun Peng, Martin Andreoni, Willian T. Lunardi, Wenke Lee

*CVPR, Workshop on Visual Anomaly and Novelty Detection, 2024*

 [PDF](#)

### **ClickDiffusion: Harnessing LLMs for Interactive Precise Image Editing**

Alec Helbling, Seongmin Lee, Polo Chau

*CVPR, Workshop on AI for Content Creation, 2024*

 [PDF](#)

### **LLM Attributor: Interactive Visual Attribution for LLM Generation**

Seongmin Lee, Zijie J Wang, Aishwarya Chakravarthy, Alec Helbling,  
ShengYun Peng, Mansi Phute, Duen Horng Chau, Minsuk Kahng  
*IEEE VIS: Visualization and Visual Analytics, 2024*



### **Mobile Fitting Room: On-device Virtual Try-on via Diffusion Models**

Justin Blalock, David Munechika, Harsha Karanth, Alec Helbling, Pratham  
Mehta, Seongmin Lee, Duen Horng Chau  
*CVPR 2024, Workshop on Virtual Try-on, 2024*

**Invited Talk**



### **ObjectComposer: Consistent Generation of Multiple Objects Without Fine-tuning**

Alec Helbling, Evan Montoya, Duen Horng (Polo) Chau  
*Preprint, 2023*



### **LLM Self Defense: By Self Examination, LLMs Know They Are Being Tricked**

Alec Helbling, Mansi Phute, Matthew Hull, Duen Horng (Polo) Chau  
*Preprint, 2023*



### **Manifold Contrastive Learning with Variational Lie Group Operators**

Kion Fallah, Alec Helbling, Kyle A. Johnsen, Christopher John Rozell  
*Preprint, 2023*



### **PrefGen: Preference Guided Image Generation with Relative Attributes**

Alec Helbling, Christopher John Rozell, Matthew O'Shaughnessy, Kion Fallah  
*Preprint, 2023*



### **Oracle Guided Image Synthesis with Relative Queries**

Alec Helbling, Christopher John Rozell, Matthew O'Shaughnessy, Kion Fallah  
*International Conference on Learning Representations, Workshop on Deep Generative Models for Highly Structured Data, 2022*

## Visualizing Convolutional Neural Network Protein-Ligand Scoring

Joshua Hochuli, Alec Helbling, Tamar Skaist, Matthew Ragoza, David Ryan Koes

*Journal of Molecular Graphics and Modeling, 2018*

 PDF

## 3Dmol.js

Alec Helbling, David Koes

*Open-source Project, 2024*

 Code (917 )

# Mentoring

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Fall 2023 **Evan Montoya** at Georgia Tech

*BS in Computer Science, Georgia Institute of Technology*

Methods for enabling compositionality of text-to-image generation systems.

Spring 2024 **Aeree Cho, Grace Kim, Alexander Karpekov** at Georgia Tech

*MS in CS, BS in CS, MS in CS, Georgia Tech*

Transformer Explainer: Interactive Learning of Text-Generative models

# References

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**Polo Chau**, Associate Professor

*School of Computational Science and Engineering*

Georgia Institute of Technology, Atlanta, GA, USA

 <https://cc.gatech.edu/~dchau/>

**Chris Rozell**, Professor

*School of Electrical & Computer Engineering*

Georgia Institute of Technology, Atlanta, GA, USA

 <https://siplab.gatech.edu/rozell.html>

**Lukas Mandrake**, Senior Researcher and Group Supervisor

*Machine Learning and Instrument Autonomy Group*

NASA Jet Propulsion Lab, Pasadena, CA, USA

 <https://ml.jpl.nasa.gov/people/mandrake/mandrake.html>

**Achille Fokoue**, Distinguished Research Staff Member

*Foundations of AI Reasoning Group*

IBM Research, Yorktown Heights, NY, USA

 <https://researcher.watson.ibm.com/researcher/view.php?person=us-achille>